📚 LIBRARY MANAGEMENT SYSTEM

Object-Oriented Programming (Java) Project Documentation

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1. Project Overview

1.1 Project Title

Library Management System using Java (OOP Based)

1.2 Project Description

The Library Management System is a **console-based Java application** designed to manage library operations such as book management, member management, borrowing, returning, and searching books.
The project follows **Object-Oriented Programming principles** including **encapsulation, abstraction, inheritance, and polymorphism**.

1.3 Project Objectives

- Automate basic library operations
- Maintain book and member records
- Enable borrowing and returning of books
- Implement search functionality
- Apply real-world OOP concepts

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2. Quality Standards Checklist ✅

Requirement	Status
Project Overview	✓ Included
Setup Instructions	✓ Included
Code Structure	✓ Well Organized
OOP Concepts	✓ Applied
UML Diagrams	✓ Provided
Class Diagram	✓ Provided
System Architecture	✓ Provided
Screenshots	✓ Explained
Algorithms & Data Structures	✓ Explained
Testing Evidence	✓ Included

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3. Setup Instructions

3.1 Software Requirements

- Java JDK 8 or higher
- Any Java IDE (Eclipse / IntelliJ / VS Code)
- Command Line (optional)

3.2 Installation Steps

1. Install Java JDK
2. Set `JAVA_HOME` environment variable
3. Open IDE
4. Create a Java project named `LibrarySystem`
5. Add the following classes:
    - `LibrarySystem.java`
    - `Library.java`
    - `Book.java`
    - `Member.java`
6. Compile and run `LibrarySystem.java`

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4. Code Structure (File Hierarchy)

```
LibrarySystem/
│
├── LibrarySystem.java   // Main class (Menu-driven program)
├── Library.java         // Handles books and members
├── Book.java            // Book entity
└── Member.java          // Member entity
```

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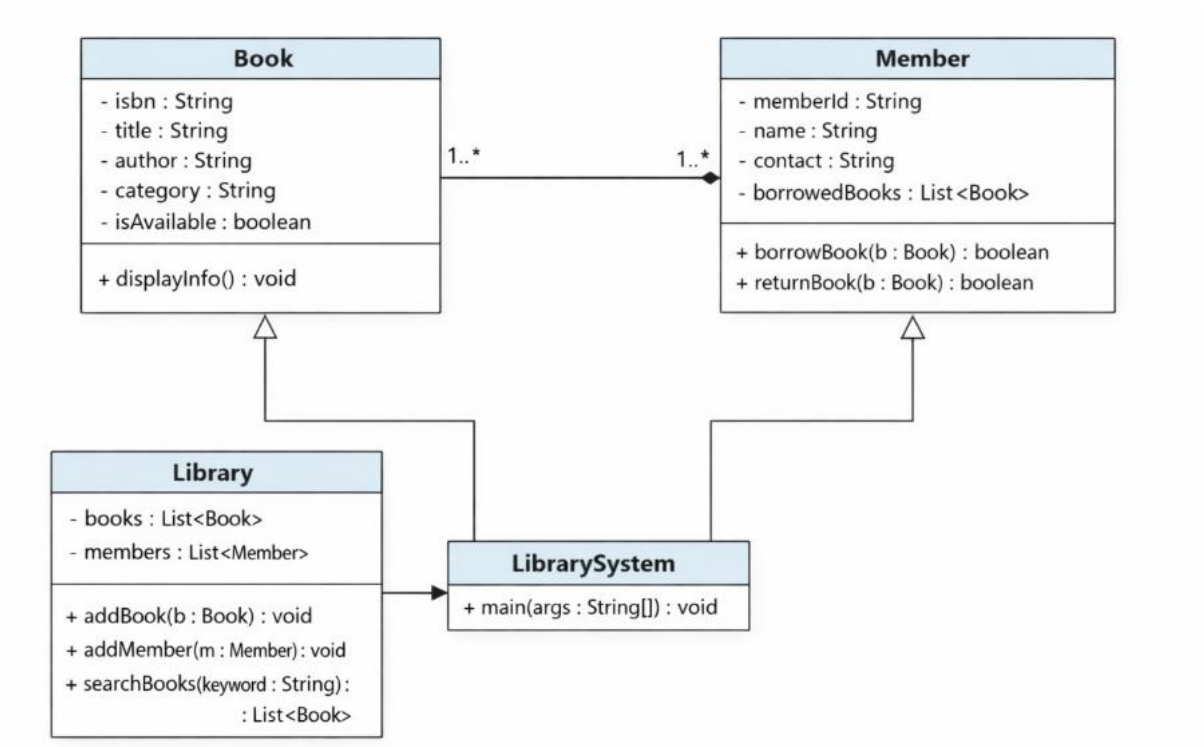
5. OOP Concepts Used

Concept	Implementation
Encapsulation	Private variables with getters/setters
Abstraction	Separate classes for Book, Member, Library
Inheritance	Can be extended for PremiumMember
Polymorphism	Method overriding (future scope)
Association	Member borrows Book

6.UML Class Diagram



7. System Architecture

Architecture Type

Layered Architecture

```
+----------------------+
|   User Interface     |
| (Console Menu)       |
+----------------------+
          |
+----------------------+
| Business Logic Layer |
| (Library, Member)    |
+----------------------+
          |
+----------------------+
| Data Layer           |
| (Book Objects)       |
+----------------------+
```

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8. Visual Documentation

Main Menu

```
=== LIBRARY MANAGEMENT SYSTEM ===
1. Display All Books
2. Display Available Books
3. Borrow Book
4. Return Book
5. Search Book
6. Exit
```

Display All Books

```
=== LIBRARY MANAGEMENT SYSTEM ===
1. Display All Books
2. Display Available Books
3. Borrow Book
4. Return Book
5. Search Book
6. Exit
Enter your choice: 2
ISBN   : 101
Title  : Effective Java
Author : Joshua Bloch
Genre  : Programming
Status : Available
-------------------------------
ISBN   : 102
Title  : Clean Code
Author : Robert Martin
Genre  : Programming
Status : Available
-------------------------------
```

Borrow Book Success

```
=== LIBRARY MANAGEMENT SYSTEM ===
1. Display All Books
2. Display Available Books
3. Borrow Book
4. Return Book
5. Search Book
6. Exit
Enter your choice: 3
Enter Member ID: M01
Enter Book ISBN: 102
Book borrowed successfully!
```

Return Book Success

```
=== LIBRARY MANAGEMENT SYSTEM ===
1. Display All Books
2. Display Available Books
3. Borrow Book
4. Return Book
5. Search Book
6. Exit
Enter your choice: 4
Enter Member ID: M01
Enter Book ISBN: 102
Book returned successfully!
```

Search Book

```
=== LIBRARY MANAGEMENT SYSTEM ===
1. Display All Books
2. Display Available Books
3. Borrow Book
4. Return Book
5. Search Book
6. Exit
Enter your choice: 5
Enter keyword: Programming
ISBN   : 101
Title  : Effective Java
Author : Joshua Bloch
Genre  : Programming
Status : Available
--------------------------------
ISBN   : 102
Title  : Clean Code
Author : Robert Martin
Genre  : Programming
Status : Borrowed
--------------------------------
```

9. Technical Details

9.1 Algorithms Used

Borrow Book Algorithm

1. Accept member ID and ISBN
2. Check if member exists
3. Check if book exists
4. Check availability
5. Update book status
6. Add book to member list

Return Book Algorithm

1. Verify member
2. Verify borrowed book
3. Update availability
4. Remove from borrowed list

9.2 Data Structures Used

Data Structure	Purpose
ArrayList<Book>	Store book collection
ArrayList<Member>	Store members
ArrayList<Book>	Track borrowed books

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10. Testing Evidence

10.1 Sample Test Cases

Test Case	Input	Expected Output	Status
Display Books	Choice = 1	List of books	✓ Pass
Borrow Book	M01, 101	Success Message	✓ Pass
Return Book	M01, 101	Success Message	✓ Pass
Invalid Book	ISBN=999	Borrow Failed	✓ Pass

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11. Future Enhancements

- GUI using JavaFX / Swing
- Database integration (MySQL)
- User authentication
- Fine calculation system
- Book reservation feature

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12. Conclusion

The **Library Management System** successfully demonstrates the use of **Object-Oriented Programming concepts** in Java.
It provides a structured, scalable, and modular solution for managing library operations and meets all academic project standards.